

Database Schema & Data Dictionary

InfusioTech Restaurant Ordering SaaS | Version 1.0 | 2 October 2026

Purpose

Provides a logical data model for implementation. Exact column types and indexes should be finalized during migrations.

Core Entities

- tenants: id, name, slug, status, timezone, contact details, branding, settings, created_at.
- users: id (auth identity), display_name, email/phone verification metadata, created_at.
- tenant_memberships: id, tenant_id, user_id, status, role_id, invited_by, joined_at; unique active membership per user/tenant.
- roles: id, tenant_id nullable for system templates, name, is_system_role.
- permissions: id, key, description; role_permissions: role_id, permission_id.
- user_permission_overrides: membership_id, permission_id, effect (grant/deny), updated_by; protect owner-only permission keys.
- staff_invitations: id, tenant_id, contact, role_id, token_hash, expiry, status, invited_by.
- subscriptions: id, tenant_id, plan_id, status, period_start/end, gateway_reference; plans and plan_entitlements hold feature limits.

Restaurant Operations

- branches (future): id, tenant_id, name, address, timezone, status.
- restaurant_tables: id, tenant_id, branch_id nullable, label, capacity, status.
- qr_assets: id, tenant_id, table_id, public_token_hash, asset_type, status, created_at.
- menu_categories: id, tenant_id, name, sort_order, is_active.
- menu_items: id, tenant_id, category_id, name, description, price_minor, currency, image_path, availability, tax_code.
- item_variants and item_addons: tenant-scoped option groups and options with price deltas stored in minor currency units.

Commerce & Customer

- customers: id, tenant_id, name/contact fields where provided, consent metadata, created_at.
- orders: id, tenant_id, table_id, customer_id nullable, order_number, order_status, payment_status, currency, subtotal_minor, tax_minor, discount_minor, total_minor, created_at.
- order_items: immutable snapshot of item name, variant/add-ons, unit price, quantity, tax and line total at order time.
- payments: id, tenant_id, order_id, provider, provider_reference, amount_minor, currency, status, verified_at.
- refunds: id, tenant_id, payment_id, amount_minor, reason, status, provider_reference.
- audit_events: id, tenant_id nullable, actor_user_id, action, entity_type/id, before/after summary, timestamp, request_id.

Data Rules

- Use UUIDs or another non-guessable primary key; store money as integer minor units plus currency.
- Use foreign keys and tenant-consistency constraints where possible; index tenant_id with common filters.
- Soft-disable operational entities where history must remain; do not mutate historical order snapshots when menu changes.
- Never store raw payment card data or invitation secrets; store only token hashes.
- RLS policies must scope all tenant records through active tenant_memberships; service-role credentials stay server-side.

ER Relationship Summary

- Tenant 1:N memberships, branches, tables, menu categories/items, orders, customers, subscriptions and audit events.
- Membership N:1 user and role; membership 1:N permission overrides.
- Order 1:N order_items, payments and refunds (through payments). Table 1:N orders.
- Menu item references category and option groups; order_items retain snapshots rather than relying on current menu values.

Migration & Validation

- Version all schema changes; apply migrations through CI/CD and review destructive changes.
- Seed system roles and permission keys through repeatable migrations.
- Test tenant A/B isolation, invitation expiry, role changes, deletion/retention and payment webhook idempotency.